

164. If $2 = x + \frac{1}{1 + \frac{1}{3 + \frac{1}{4}}}$, then the value of x is

- (a) $\frac{12}{17}$ (b) $\frac{13}{17}$
(c) $\frac{18}{17}$ (d) $\frac{21}{17}$

165. If $\frac{2 + \frac{1}{3 + \frac{4}{5}}}{2 + \frac{1}{3 + \frac{1}{1 + \frac{1}{4}}}} = x$, then the value of x is

- (a) $\frac{1}{7}$ (b) $\frac{3}{7}$
(c) 1 (d) $\frac{8}{7}$

166. $8 - 8 \times \frac{2\frac{1}{5} - 1\frac{2}{7}}{2 - \frac{1}{6 - \frac{1}{6}}}$ is equal to

- (a) 2 (b) 4
(c) 6 (d) 8

167. $\frac{2}{2 + \frac{2}{3 + \frac{2}{3 + \frac{2}{3}}}} \times 0.39$ is simplified to

- (a) $\frac{1}{3}$ (b) 2
(c) 6 (d) None of these

168. Simplify: $\frac{1}{1 + \frac{\frac{2}{3}}{1 + \frac{\frac{8}{9}}{1 + \frac{2}{3} + \frac{9}{1 - \frac{2}{3}}}}}$

- (a) $\frac{11}{13}$ (b) $\frac{13}{15}$
(c) $\frac{13}{11}$ (d) $\frac{15}{13}$

169. If $\frac{2}{2 + \frac{2}{2 + \frac{2}{x}}} = 3$, then x is equal to

- (a) $-\frac{4}{7}$ (b) $\frac{4}{7}$
(c) $-\frac{4}{9}$ (d) $\frac{4}{9}$

170. If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y, z are natural

numbers, then x, y, z are

- (a) 1, 2, 5 (b) 1, 5, 2
(c) 5, 2, 11 (d) 11, 2, 5

171. Let $x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots \infty}}}$. Which of the following

is correct?

- (a) $x^2 + x + 1 = 0$ (b) $x^2 - x + 1 = 0$
(c) $x^2 + x - 1 = 0$ (d) $x^2 - x - 1 = 0$

172. Let $P = \frac{1}{1 + \frac{1}{1 + \frac{1}{3 + \frac{1}{4 + \dots}}}}$ and $Q = \frac{1}{2 + \frac{1}{3 + \frac{1}{4 + \frac{1}{\dots}}}}$.

Then, $P + Q$ when calculated, gives

- (a) 1 (b) 2
(c) 3 (d) 4

173. $\frac{\left[\left(1 + \frac{1}{10 + \frac{1}{10}} \right) \left(1 + \frac{1}{10 + \frac{1}{10}} \right) - \left(1 - \frac{1}{10 + \frac{1}{10}} \right) \left(1 - \frac{1}{10 + \frac{1}{10}} \right) \right]}{\left[\left(1 + \frac{1}{10 + \frac{1}{10}} \right) + \left(1 - \frac{1}{10 + \frac{1}{10}} \right) \right]}$

simplifies to

- (a) $\frac{20}{101}$ (b) $\frac{90}{101}$
(c) $\frac{100}{101}$ (d) $\frac{101}{100}$

174. Which of the following values of x and y satisfy the following equations I and II?

- I. $3x + y = 19$ II. $x - y = 9$
(a) -7, -2 (b) -7, 2
(c) 7, -2 (d) 7, 2

175. If $4x = p(x + 3) + q(x - 1)$ is an identity, then the values of p and q are

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- (a) 1, -3 (b) 1, 3
(c) 1, 1 (d) 3, 1